

Worksheet 4C: Advanced Rules — Derivatives and Their Integral Inverses

AIML Foundations Mathematics
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What This Worksheet Is About Each derivative rule has an integral counterpart: - **Product Rule** $\leftrightarrow$ **Integration by Parts** - **Chain Rule** $\leftrightarrow$ **Substitution (u-sub)** These are inverse operations!

Part A: The Product Rule (12 problems)

$$\frac{d}{dx}[u \cdot v] = u' \cdot v + u \cdot v'$$

Find the derivative:

1. $f(x) = x^2(x + 1)$
2. $g(t) = t^3(t - 4)$
3. $h(p) = (p + 2)(p - 5)$
4. $y = (2x + 1)(3x - 2)$
5. $f(m) = m^2(m^2 + 3m - 1)$
6. $g(r) = (r^2 - 1)(r^2 + 1)$
7. Verify problem 3 by expanding first.
8. Verify problem 6 by simplifying first: $(r^2 - 1)(r^2 + 1) = ?$ **With Chain Rule:**
9. $h(x) = x^2(x + 1)^3$
10. $y = (t + 1)^2(t - 1)^3$
11. $f(k) = k(2k + 1)^4$
12. $g(n) = (n^2 + 1)^2(n - 1)$

Part B: Integration by Parts — Reversing Product Rule (10 problems)

$$\int u \, dv = uv - \int v \, du$$

Connection: If $\frac{d}{dx}[f \cdot g] = f'g + fg'$, then $\int(f'g + fg')dx = fg + C$

13. The derivative of $x^2(x+1)$ is $3x^2+2x$. Therefore: $\int(3x^2+2x)dx =$ _____
14. The derivative of $(x-1)(x+3)$ is $2x+2$. Therefore: $\int(2x+2)dx =$ _____
15. Find $\frac{d}{dx}[(2x+1)(x-2)]$. Then verify $\int$ [your answer] dx gives back $(2x+1)(x-2)+C$.
16. Start with $f(x) = x(x+1)^2$. Differentiate, then integrate your answer. **Direct Integration (for comparison):**
17. $\int(3x^2 + 2x)dx =$
18. $\int(2x + 2)dx =$
19. Compare: Are problems 13 & 17 the same? Are 14 & 18 the same (up to constants)?
20. Explain why integration is "harder" than differentiation: - Differentiation: clear rules always work - Integration: may require _____ or may not have closed form **Preview: Integration by Parts with transcendentals (Worksheet 4D)**
21. We'll need by parts for $\int x \cdot e^x dx$ because we can't _____.
22. And for $\int x \cdot \sin(x)dx$ because _____.

Part C: The Quotient Rule (10 problems)

$$\frac{d}{dx} \left[\frac{u}{v} \right] = \frac{u'v - uv'}{v^2}$$

Find the derivative:

23. $f(x) = \frac{x}{x+2}$
24. $g(t) = \frac{t^2}{t-1}$
25. $h(p) = \frac{p+1}{p-1}$
26. $y = \frac{x^2+1}{x}$
27. $f(m) = \frac{1}{m^2+1}$
28. $g(r) = \frac{r^3}{r^2+1}$ **Quotient Rule vs. Rewriting:**
29. Find $\frac{d}{dx} \left[\frac{1}{x^3} \right]$ two ways:
 - (a) Quotient rule
 - (b) Rewrite as x^{-3}
30. Find $\frac{d}{dx} \left[\frac{x^4+2x^2}{x^2} \right]$ by simplifying first.
31. Simplify first, then differentiate: $\frac{x^2-1}{x+1}$
32. When MUST you use quotient rule? (When denominator has _____ that don't cancel)

Part D: The Chain Rule (16 problems)

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

"Derivative of outside $\times$ derivative of inside"

Find the derivative:

33. $f(x) = (x + 3)^4$

34. $g(t) = (2t - 1)^5$

35. $h(p) = (p^2 + 1)^3$

36. $y = (3x^2 - x)^4$

37. $f(m) = (m^3 + m)^2$

38. $g(r) = (1 - r^2)^6$ **Negative and Fractional Powers:**

39. $h(x) = (x + 1)^{-1}$

40. $y = (t^2 + 4)^{-2}$

41. $f(k) = \sqrt{k + 5}$

42. $g(n) = \sqrt{n^2 + 1}$ **Verify by Expanding:**

43. $(x + 2)^3$: Use chain rule, then expand and differentiate. Same answer?

44. $(2x - 1)^2$: Both methods. **Combining Rules:**

45. $f(x) = x^2(x + 1)^3$ (product + chain)

46. $g(t) = \frac{t}{(t+1)^2}$ (quotient + chain)

47. $h(x) = \sqrt{x(x + 4)}$ (chain with product inside)

48. $y = \left(\frac{x}{x+1}\right)^3$ (chain on quotient)

Part E: Substitution — Reversing the Chain Rule (16 problems)

$$\int f(g(x)) \cdot g'(x) dx = F(g(x)) + C$$

**Let $u = g(x)$, then $du = g'(x)dx$ **

Perfect Cases:

49. $\int 3x^2(x^3 + 1)^4 dx$ Let $u = x^3 + 1$, $du = 3x^2 dx = \int u^4 du =$

50. $\int 2t(t^2 - 3)^5 dt$
51. $\int 4p^3(p^4 + 2)^{-2} dp$
52. $\int \frac{6x^2}{\sqrt{x^3+1}} dx$
53. $\int (2m + 1)(m^2 + m)^3 dm$ **Adjusting Constants:**
54. $\int x(x^2 + 1)^3 dx$ We need $2x dx$ but have $x dx$. So: $x dx = \frac{1}{2}(2x dx) = \frac{1}{2} du = \frac{1}{2} \int u^3 du =$
55. $\int t(t^2 + 5)^4 dt$
56. $\int \frac{x}{\sqrt{x^2+4}} dx$
57. $\int p^2(p^3 - 1)^2 dp$ **Verify by Differentiating:**
58. Check problem 49: Differentiate $\frac{(x^3+1)^5}{5}$ using chain rule.
59. Check problem 54: Differentiate $\frac{(x^2+1)^4}{8}$. **Connection to Chain Rule:**
60. Chain rule: $\frac{d}{dx}[(x^2+1)^5] = 10x(x^2+1)^4$ Therefore: $\int 10x(x^2+1)^4 dx =$ _____
61. Chain rule: $\frac{d}{dx}[(t^3-2)^4] = 12t^2(t^3-2)^3$ Therefore: $\int 12t^2(t^3-2)^3 dt =$ _____
62. Explain: Substitution is "chain rule backwards." **More Practice:**
63. $\int (3x - 2)^4 dx$
64. $\int (5t + 1)^{-3} dt$

Part F: Combining Everything (10 problems)

Differentiate:

65. $\frac{x^2}{(x+1)^3}$
66. $x(x^2 + 1)^{1/2}$
67. $\frac{(x-1)^2}{(x+1)^2}$ **Integrate:**
68. $\int \frac{2x}{(x^2+1)^3} dx$
69. $\int \frac{x}{\sqrt{x^2+9}} dx$
70. $\int (x + 1)(x^2 + 2x)^4 dx$ *Hint: $\frac{d}{dx}(x^2 + 2x) = 2x + 2 = 2(x + 1)$ * **Round-Trips:**
71. Start with $(x^2 + 1)^3$. Differentiate, then integrate back.
72. Start with $x(x + 1)^2$. Differentiate, then integrate back.
73. Start with $\frac{x}{x+1}$. Differentiate, then integrate back.
74. Why does "differentiate then integrate" give $f(x) + C$, but "integrate then differentiate" gives exactly $f(x)$?